

Pratiksha K.P

Software Engineer | Full-Stack Development & Machine Learning
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EDUCATION

B.E. Computer Science (Majors) Business Systems (Minors) BMS College of Engineering, Bengaluru	Aug 2023 – May 2027 CGPA: 9.2 / 10.0
PU I & II — Science (PCM) Deeksha Center for Learning, Bengaluru (Karnataka Board)	2021 – 2023 93.5% 98% aggregate in PCM
Class 10 — ICSE Ryan International School	2021 93.16%

EXPERIENCE

Research Intern — IoT & Indoor Localization Samsung R&D Institute, Bengaluru	Jan 2026 – Present
- Developed RSSI-based Android data collection app from scratch as validation baseline before migrating to WiFi RTT (IEEE 802.11mc FTM), enabling systematic ranging accuracy comparison across 3+ residential layouts. - Conducted literature survey on WiFi RTT indoor positioning (EKF-RW/EKF-SH, distance-dependent measurement models) based on Samsung Research America's IEEE Access publication. - Authored business case study mapping WiFi RTT localization to Samsung SmartThings expansion across residential and enterprise deployments.	

PROJECTS

Mini Jenkins — CI/CD Pipeline Simulator Node.js · Express · GitHub Webhooks · ngrok · Vanilla JS github.com/Pratiksha-KP/our-jenkins	- Built a CI/CD pipeline simulator modelling Jenkins: priority-based job scheduler scoring jobs by branch (main > staging > develop) and language, with a typed worker pool (Node, Python, Java, Generic) assigning jobs via language-affinity matching with fallback. - Integrated real GitHub webhook events via ngrok tunneling — commits to linked repos trigger live pipeline runs; implemented a Jenkinsfile stage parser to extract and track Build/Test/Deploy stages per job. - Developed a real-time dashboard (vanilla JS, 3-second polling) showing live job queue, worker states, activity log, and job lifecycle (QUEUED → RUNNING → SUCCESS/FAILED) with priority badges and duration tracking.
UberFriends — Ride Coordination Platform Node.js · Express · PostgreSQL · WebSockets · JWT · Google Places API github.com/ishani4208/uberfriends2.0	- Architected entire backend using RESTful microservices design: Haversine-based driver matching with ACID-compliant transactional SQL preventing race conditions across concurrent ride requests. - Built WebSocket notification system (~20 concurrent users, sub-200ms latency); JWT + bcrypt auth; unit tests for core matching and fare logic. - Integrated Google Places Autocomplete and distance-based dynamic fare calculation across 10+ test destinations.
BiteDash — Real-Time Canteen Ordering System React · Node.js · PostgreSQL · Twilio · JWT github.com/Pratiksha-KP/Canteen_vendor-management	- Designed PostgreSQL schema with triggers for auto queue management; Twilio SMS notifications; 50+ concurrent orders across Agile sprint-based development cycles. - Built React vendor dashboard with 4 views (orders, new order, menu management, analytics); implemented 5-second auto-refresh polling for live order status tracking. - Integrated full Twilio SMS lifecycle (order placed, preparing, ready, cancellation via DROP keyword webhook); analytics view surfacing daily revenue, top 5 items by sales, and order status breakdown.

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript
Backend / DB: Node.js, Express.js, REST APIs, WebSockets, JWT, PostgreSQL, SQL, Microservices
Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS
Tools & Platforms: Git, GitHub, Twilio, Android SDK, Google Cloud Console, Postman, CI/CD
Coursework: DSA, DBMS, OS, Computer Networks, Cryptography, Deep Learning

CERTIFICATIONS

- Nutanix Certified Associate 6 (NCA 6) — Multicloud Infrastructure | Issued Apr 2026
- NPTel — Human Computer Interaction (Elite, Top 5% of 23,139 candidates, 98% score) | Jan–Apr 2026
- Machine Learning with Python — IBM / Coursera | Issued Nov 2025

LEADERSHIP & ACTIVITIES

Club Coordinator, Google Developer Groups on Campus BMSCE — UTSAV 2026, Inter-Collegiate Techno Cultural Fest | **Management Lead, Google Developer Groups on Campus BMSCE** | **IEEE Computer Society** | **Carnatic Music:** Junior-level certified practitioner and currently learning Senior level